

Method	Type	How a head is scored	How heads are removed	Exact count	Never broke	Backups
Collateral rule	ours	$\min_W L_{-h} - \min_W L$ loss after removing h , read-out re-fitted	during training, from scratch; price, no target; gradual, reopens	29/30	30/30	predicted
Michel et al. 2019	published	$ \partial L / \partial g_h $ gradient of the loss w.r.t. head gate	one at a time, while the loss stays below the solved bar	28/30	0/30	no
Voita et al. 2019	published	learned gate g_h : $L + \lambda \sum_h P(g_h \neq 0)$ hard-concrete (L0) penalty	learned during training	9/30*	0/30*	no
Same score, all at once	control	$\min_W L_{-h} - \min_W L$ (the OBS / ZipLM way)	all at once after the task is solved, then fine-tuned	24/30	0/30	no
Ordinary importance	control	$L_{-h}(W) - L(W)$ loss after removing h , no re-fit	our rule, this score instead	2/30	30/30	no
Random choice	control	head picked at random	our rule, random head	30/30	6/30	no

Our score in closed form: $v_h = \text{tr}(W_h^\top [M_{hh}]^{-1} W_h) / d$, $M = (Z^\top Z / n)^{-1}$ (Z : the heads' outputs). This is the Optimal Brain Surgeon / ZipLM score $\frac{1}{2} w_q^\top ([H^{-1}]_{qq})^{-1} w_q$ with $H = 2 Z^\top Z / n$, the Hessian of the read-out. The score is not new; what is new is using it during training, with a price instead of a target size, gradually.

Exact count: heads left = heads the task needs. Never broke: once solved, the loss never went back above the solved bar.

Backups: how many spare heads are kept when heads can fail, predicted before the runs. *best of 3 penalty strengths. Every method on the same 30 runs (tasks needing 2, 4 and 8 heads, 10